

# Anushri Dixit

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## Education

- 2017 – 2023 **Ph.D., California Institute of Technology** in Control and Dynamical Systems,  
Thesis title: *Risk-Aware Planning and Control in Extreme Environments*.
- 2013 – 2017 **B.Sc., Georgia Institute of Technology** in Electrical Engineering, Highest Honors,  
Thesis title: *System Dynamics-Based Mapping for Closed Loop Control*.

## Research Interests

Stochastic motion planning, robotics, artificial intelligence, control theory, safety-critical systems, probability theory.

## Appointments

- July 2024 – **Assistant Professor, University of California, Los Angeles**  
Tenure-Track Faculty in the Department of Mechanical & Aerospace Engineering
- March 2023 – May 2024 **Postdoctoral Researcher, Princeton University**

## Teaching Experience

- 2024– **UCLA courses.**  
*MAE 171A* - Undergraduate class on Introduction to Feedback and Control Systems.  
*MAE 271D* - Graduate class on Special Topics on Dynamic Systems Control.  
*MAE 82* - Undergraduate class on Mathematics of Engineering.
- 2021–2022 **Graduate Teaching Assistant, Advanced Robotics (ME 234a and ME 243b).**  
Helped design part of the course and taught 3 weeks (9+ hours) of lectures on MPC.  
Held laboratory sessions and office hours for students and graded assignments.
- 2019 **Summer Undergraduate Research Fellowships (SURF) program Mentor.**  
Mentored undergraduate students to build an autonomous RC car for the DARPA Subterranean Challenge.
- 2018 – 2019 **Graduate Teaching Assistant, Robotics (ME 133a and ME 134b).**  
Held laboratory sessions and office hours for 30+ students and graded assignments.
- 2015 – 2017 **Undergraduate Teaching Assistant, Differential Equations (MATH 2552).**  
Held recitation sessions for 30+ students and received positive feedback from students and lecturer. Proctored and graded quizzes as well as exams.

## Awards and Fellowships

- 2025 **UCLA COR Research Allowance Program**
- 2023 **Best Student Paper Award at the Conference on Robot Learning (as a co-author) [C5]**
- 2022 **Outstanding Service as a Reviewer for IEEE Control System Letters**  
**Outstanding Student Paper Award at IEEE Conference on Decision and Control (with Skylar Wei) [J8]**  
**University of Chicago Rising Star in Data Science**  
**Southern California Robotics Symposium Rising Star**
- 2021 **DE Shaw Zenith Fellowship**

## Invited Talks

- February 2025 **Making Robots Trustworthy: Understanding Risk and Uncertainty for Safe Autonomy**  
Stanford University, *Stanford Robotics and Autonomous Systems Seminar*
- March 2025 **Planning with Confidence: Uncertainty Quantification for Safety-Critical Tasks**  
Carnegie Mellon University, *Guest Lecture - Embodied AI Safety*
- October 2024 **University of Southern California, Guest Lecture - Foundations and Algorithms for Intelligent Autonomous Systems**
- February 2024 **University of California, San Diego, Guest Lecture - Safety for Autonomous Systems**

## Invited Talks (continued)

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| July 2024      | <b>Robots that know when they don't know: Uncertainty Quantification for LLM-based planners</b><br>Robotics: Science and Systems, <i>Workshop on Robots that help and ask for help</i>      |
| March 2024     | <b>Robot Safety and Generalization in the Era of Foundation Models</b><br>University of Maryland, <i>Microsoft Future Leaders in Robotics and AI Seminar Series</i>                         |
| March 2024     | <b>Perceive with Confidence: Statistical Safety Assurances for Vision-Based Navigation</b><br><i>Talking Robotics</i>                                                                       |
| November 2023  | <b>Robots that Ask for Help: Uncertainty Alignment for LLM Planners</b><br>Princeton University, <i>Guest Lecture - Inference in Action: Probabilistic Topics in Reinforcement Learning</i> |
| August 2023    | <b>Risk-Aware Control and Planning in Unstructured Environments</b><br>University of Washington, Seattle, <i>Control-X Seminar</i>                                                          |
| November 2022  | University of Chicago, <i>Rising Stars in Data Science</i>                                                                                                                                  |
| September 2022 | University of California, Los Angeles, <i>Southern California Robotics Symposium</i>                                                                                                        |
| December 2022  | <b>Distributionally Robust Model Predictive Control With Total Variation Distance</b><br>Conference on Decision and Control, <i>Invited Session</i>                                         |
| December 2021  | <b>Risk-Averse Stochastic Shortest Path Planning</b><br>Conference on Decision and Control, <i>Invited Session</i>                                                                          |
| June 2021      | <b>Risk-Sensitive Motion Planning using Entropic Value-at-Risk</b><br>European Control Conference                                                                                           |

## Service

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| 2025           | Co-organizer of 2 <sup>nd</sup> Workshop on Out-of-Distribution Generalization in Robotics at Robotics: Science and Systems.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 2024 - Present | Co-organizer of the UCLA MAE Departmental Seminar.<br>Mentor of the UCLA student organization competing in the University Rover Competition (URC@UCLA).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| 2023           | Co-organizer of the Princeton Robotics Seminar.<br>Co-organizer of 2023 Workshop on Out-of-Distribution Generalization in Robotics at the Conference on Robot Learning (CoRL).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| 2019 – Present | <b>Reviewer (Journals):</b> Automatica, Field Robotics, IEEE Control Systems Letters (L-CSS), Journal of Aerospace Information Systems, IEEE Transactions on Robotics (T-RO).<br><b>Reviewer (Conferences):</b> Conference on Decision and Control (CDC), American Control Conference (ACC), IEEE International Conference on Robotics and Automation (ICRA), IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), Learning for Dynamics & Control Conference (L4DC), Robotics: Science and Systems (RSS).<br><b>Reviewer (Workshops):</b> Bridging the Lab-to-Real Gap (ICRA 2023), Out-of-Distribution Generalization in Robotics (CoRL 2023). |
| 2024 – Present | <b>Reviewer (Proposals):</b> National Science Foundation (Information and Intelligent Systems), Air Force Office of Scientific Research, NASA Space Technology Graduate Research Opportunities.                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |

## Outreach

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| 2025 - Present | <b>Volunteer, Robotics for All</b><br>Helping create a new virtual curriculum for robotics and control using Scratch programming to improve access to robotics education across middle schools.                    |
| Summer 2023    | <b>Mentor, Asian American Academy of Science and Engineering (AAASE) Summer Academy</b><br>Mentored five high-school students for a project on search and rescue robotics.                                         |
| 2019 – 2021    | <b>Visiting Scientist, Caltech Center for Teaching, Learning, and Outreach</b><br>Provided hands-on science lessons at a elementary school in Pasadena as a part of a teaching program called Visiting Scientists. |
| 2015 – 2016    | <b>Chief Technical Officer, Robogals Global</b><br>Oversaw the maintenance and development of the myRobogals portal, Robogals Global website while managing a software team of 4-5 engineers.                      |

## References (\* indicates equal contribution)

### Journal Articles

- [J1] P. Akella\*, **A. Dixit\***, M. Ahmadi, L. Lindemann, M. P. Chapman, G. J. Pappas, A. D. Ames, and J. W. Burdick, "Risk-aware robotics: Tail risk measures in planning, control, and verification," *IEEE Control Systems Magazine (Accepted for publication)*, 2025.  DOI: 10.1109/MCS.2025.3577050.
- [J2] **A. Dixit\***, D. D. Fan\*, K. Otsu, S. Dey, A.-A. Agha-Mohammadi, and J. W. Burdick, "STEP: Stochastic Traversability Evaluation and Planning for Risk-Aware Navigation; Results From the DARPA Subterranean Challenge," *IEEE Transactions on Field Robotics*, vol. 2, pp. 81–99, 2025.  DOI: 10.1109/TFR.2024.3512433.
- [J3] P. Akella, **A. Dixit**, M. Ahmadi, J. W. Burdick, and A. D. Ames, "Sample-based bounds for coherent risk measures: Applications to policy synthesis and verification," *Artificial Intelligence*, 2024.  DOI: 10.48550/ARXIV.2204.09833.
- [J4] B. Morrell, K. Otsu, A. Agha, *et al.*, "An Addendum to NeBula: Toward Extending Team CoSTAR's Solution to Larger Scale Environments," *IEEE Transactions on Field Robotics*, vol. 1, pp. 476–526, 2024.  DOI: 10.1109/TFR.2024.3430891.
- [J5] **A. Dixit**, M. Ahmadi, and J. W. Burdick, "Risk-averse receding horizon motion planning for obstacle avoidance using coherent risk measures," *Artificial Intelligence*, vol. 325, p. 104 018, 2023.  DOI: 10.1016/j.artint.2023.104018.
- [J6] A. Agha, K. Otsu, B. Morrell, *et al.*, "NeBula: TEAM CoSTAR's Robotic Autonomy Solution that Won Phase II of DARPA Subterranean Challenge," *Field Robotics*, vol. 2, pp. 1432–1506, Mar. 2022.  DOI: 10.55417/fr.2022047.
- [J7] **A. Dixit**, M. Ahmadi, and J. W. Burdick, "Distributionally robust model predictive control with total variation distance," *IEEE Control Systems Letters*, vol. 6, pp. 3325–3330, 2022.  DOI: 10.1109/LCSYS.2022.3184921, **Invited Paper at CDC 2022**.
- [J8] S. X. Wei\*, **A. Dixit\***, S. Tomar, and J. W. Burdick, "Moving obstacle avoidance: A data-driven risk-aware approach," *IEEE Control Systems Letters*, vol. 7, pp. 289–294, 2022.  DOI: 10.1109/LCSYS.2022.3181191, **Outstanding Student Paper Award at CDC 2022**.

### Conference Proceedings

- [C1] Y. Zeng and **A. Dixit**, "Risk-aware reinforcement learning with bandit-based adaptation for quadrupedal locomotion," in *2026 IEEE International Conference on Robotics and Automation (ICRA)*, 2025. arXiv: 2510.14338 [cs.R0].
- [C2] **A. Dixit**, Z. Mei, M. Booker, M. Storey-Matsutani, A. Z. Ren, and A. Majumdar, "Perceive with confidence: Statistical safety assurances for navigation with learning-based perception," in *Conference on Robot Learning (CoRL)*, 2024. arXiv: 2403.08185 [cs.R0].
- [C3] A. Z. Ren, J. Clark, **A. Dixit**, M. Itkina, A. Majumdar, and D. Sadigh, "Explore until confident: Efficient exploration for embodied question answering," in *Robotics: Science and Systems*, 2024. arXiv: 2403.15941 [cs.R0].
- [C4] **A. Dixit\***, L. Lindemann\*, S. Wei, M. Cleaveland, G. J. Pappas, and J. W. Burdick, "Adaptive conformal prediction for motion planning among dynamic agents," in *Learning for Dynamics and Control (L4DC) Conference*, 2023.  [Online]. Available: <https://arxiv.org/pdf/2212.00278.pdf>.
- [C5] A. Z. Ren, **A. Dixit**, A. Bodrova, S. Singh, S. Tu, N. Brown, P. Xu, L. Takayama, F. Xia, J. Varley, Z. Xu, D. Sadigh, A. Zeng, and A. Majumdar, "Robots that ask for help: Uncertainty alignment for large language model planners," in *Conference on Robot Learning (CoRL)*, 2023. arXiv: 2307.01928 [cs.R0], **Best Student Paper Award**.
- [C6] S. Dey, D. Fan, R. Schmid, **A. Dixit**, K. Otsu, T. Touma, A. F. Schilling, and A.-a. Agha-mohammadi, "PrePARE: predictive proprioception for agile failure event detection in robotic exploration of extreme terrains," in *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2022.
- [C7] M. Ahmadi, **A. Dixit**, J. W. Burdick, and A. D. Ames, "Risk-averse stochastic shortest path planning," in *Conference on Decision and Control*, 2021.  [Online]. Available: <https://arxiv.org/abs/2103.14727>, **Invited Paper at CDC 2021**.
- [C8] **A. Dixit**, M. Ahmadi, and J. W. Burdick, "Risk-Sensitive Motion Planning using Entropic Value-at-Risk," in *European Control Conference*, 2021.  [Online]. Available: <https://arxiv.org/abs/2011.11211>.
- [C9] D. D. Fan\*, K. Otsu\*, Y. Kubo, **A. Dixit**, J. Burdick, and A.-A. Agha-Mohammadi, "STEP: Stochastic traversability evaluation and planning for safe off-road navigation," in *Robotics: Science and Systems*, 2021.  [Online]. Available: <https://arxiv.org/abs/2103.02828>.